

1D CNN — Architecture Variant Comparison

Generated 2026-05-08 13:56:26

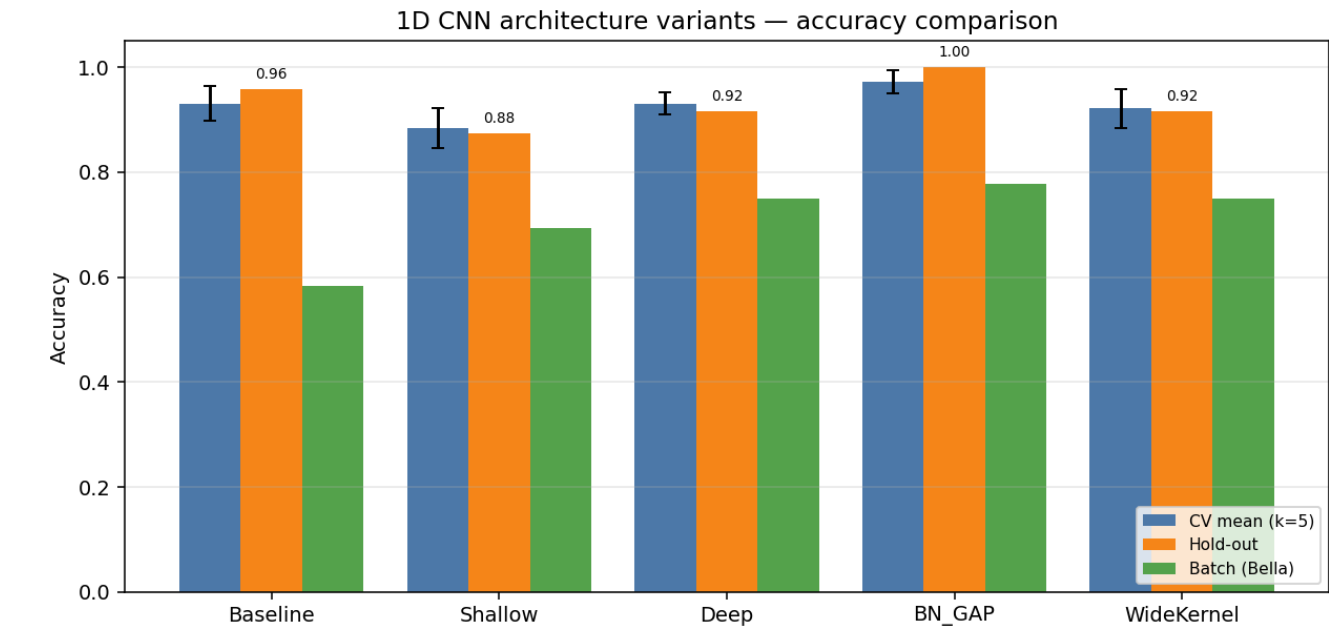
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Total trials	240 (train pool 216, hold-out 24)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

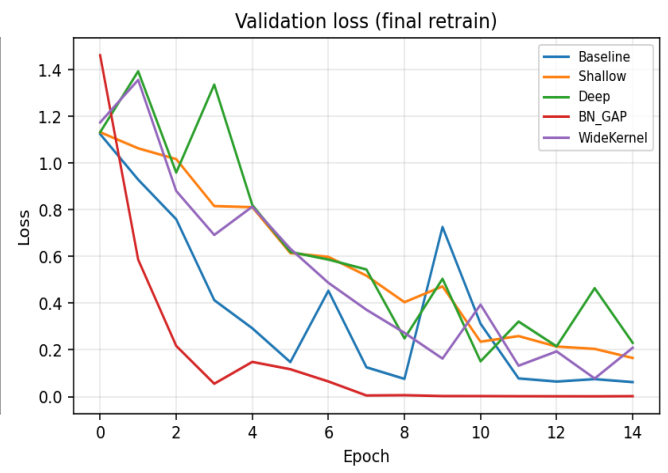
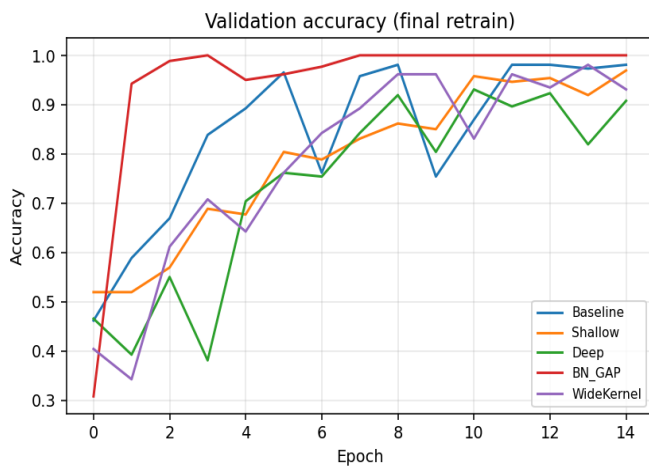
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9304	0.0330	0.9583	0.1248	0.5833	21/36
Shallow	72,487	0.8845	0.0380	0.8750	0.2424	0.6944	25/36
Deep	204,007	0.9304	0.0210	0.9167	0.3787	0.7500	27/36
BN_GAP	57,319	0.9723	0.0226	1.0000	0.0005	0.7778	28/36
WideKernel	131,687	0.9214	0.0376	0.9167	0.4640	0.7500	27/36

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9304 ± 0.0330
Hold-out test acc / loss	0.9583 · 0.1248
Batch-test acc	0.5833 (21/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	0.9767	1.0000	0.9767
3	1038	43	0.8837	0.9535	0.8837
4	1038	43	0.9070	0.9302	0.9070
5	1038	43	0.9302	0.9535	0.9302

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.667	0.800	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.964	0.952	0.951	24
weighted avg	0.969	0.958	0.957	24

Batch test per class (Bella)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	6	0.000
Double_Pistol_Recoil	4	5	0.800
Double_Raise	2	5	0.400
Double_Wiggle	0	5	0.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	21	36	0.583

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8845 ± 0.0380
Hold-out test acc / loss	0.8750 · 0.2424
Batch-test acc	0.6944 (25/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.8409	0.8864	0.8409
2	1038	43	0.9535	0.9535	0.9535
3	1038	43	0.8605	0.9070	0.8605
4	1038	43	0.8837	0.8837	0.8837
5	1038	43	0.8837	0.9302	0.8837

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.600	1.000	0.750	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.667	0.800	3
macro avg	0.914	0.857	0.848	24
weighted avg	0.917	0.875	0.863	24

Batch test per class (Bella)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	6	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	25	36	0.694

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9304 ± 0.0210
Hold-out test acc / loss	0.9167 · 0.3787
Batch-test acc	0.7500 (27/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9773	0.9545
2	1038	43	0.9302	1.0000	0.9302
3	1038	43	0.9535	0.9535	0.9535
4	1038	43	0.9070	0.9302	0.9070
5	1038	43	0.9070	0.9767	0.9070

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	24
weighted avg	0.935	0.917	0.901	24

Batch test per class (Bella)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	6	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	2	5	0.400
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	27	36	0.750

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9723 ± 0.0226
Hold-out test acc / loss	1.0000 · 0.0005
Batch-test acc	0.7778 (28/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	1.0000	1.0000	1.0000
3	1038	43	1.0000	1.0000	1.0000
4	1038	43	0.9535	0.9767	0.9535
5	1038	43	0.9535	1.0000	0.9535

Per-class hold-out metrics

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Double_Nothing	1.000	1.000	1.000	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	1.000	1.000	1.000	24
weighted avg	1.000	1.000	1.000	24

Batch test per class (Bella)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	6	0.333
Double_Pistol_Recoil	5	5	1.000
Double_Raise	3	5	0.600
Double_Wiggle	3	5	0.600
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	28	36	0.778

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9214 ± 0.0376
Hold-out test acc / loss	0.9167 · 0.4640
Batch-test acc	0.7500 (27/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9091	0.9545	0.9091
2	1038	43	0.9767	0.9767	0.9767
3	1038	43	0.9302	0.9302	0.9302
4	1038	43	0.8605	0.9070	0.8605
5	1038	43	0.9302	0.9535	0.9302

Per-class hold-out metrics

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Double_Lower	1.000	1.000	1.000	3
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Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	24
weighted avg	0.935	0.917	0.901	24

Batch test per class (Bella)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	6	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	2	5	0.400
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	27	36	0.750